Senior Engineering Leader

Majid Mehrnia

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PROFILE

Senior Engineering Leader with 12+ years of experience in pump and fluid systems development, NPI/NPD, and technology roadmapping. Expertise in engineering excellence, Design-to-Cost, VAVE, reliability engineering, and KPI-driven execution. Experienced in cross-functional collaboration and lifecycle management to deliver customer value, operational efficiency, and sustainable, profitable growth.

PROFESSIONAL EXPERIENCE

Project | September 2025 – Present

- **Pump Design & Product Development:** Conceptual design, simulation, performance evaluation, and parametric optimization of pump systems based on application requirements and fluid properties
- **Precision Fluid Handling:** Peristaltic & Diaphragm Pumps
- **High-Viscosity Applications:** Twin-screw & Progressive cavity Pumps
- **Fluid System Development & Cost Optimization:** Design improvement, performance optimization, and cost-reduction opportunities across fluid-handling systems, balancing functional requirements, efficiency and reliability

Institut für Fluidsystemtechnik, Darmstadt, Germany

R&D Engineer | February 2020 – May 2025

- Delivered VDMA FKM Project for the Pumps + Systems trade association, conducting technical reviews and engineering assessments of screw and gear pumps
- Directed the design, development & industrialization of a precision hydraulic system with dual motor-pump control for advanced metrology

Mitsubishi Heavy Industries Ltd., Tokyo, Japan

Engineering Excellence Lead | February 2018 – December 2019

- Led development of the turbocharger compressor system, optimizing casing geometry and impeller design to achieve a 19% reduction in volumetric footprint

Tadbirgaran System Ltd. (Yazd Science and Technology Park), Iran

Engineering Manager | July 2014 – September 2017

Engineering | March 2011 – June 2014

- Supported E2E NPD of Pump-as-Turbine (PAT) systems through a V-Model lifecycle, achieving 84% system efficiency while optimizing design and manufacturing cost
- Managed Product Lifecycle Management, drove Design-to-Cost and VAVE initiatives to reduce tooling and casting costs, delivering 20% ROI

EDUCATION

- PhD in Mechanical Engineering, Technische Universität Darmstadt, Germany
- MSc in Aerospace Engineering (Aerodynamics), University of Tehran, Iran
- BSc in Aerospace Engineering, Sharif University of Technology, Iran

TECHNICAL SKILLS

Engineering Management

- Technology Roadmapping
- Program Management (MSP & Power BI)
- KPI Management (LT, LCC, TCO & TTM)
- Digital Thread, PLM & SAP (e/m/s-BOM)
- PCP/PEP Gate Reviews & V-Model
- Hybrid Agile Frameworks

Systems Engineering

- MBSE & Digital Twin (Modularity)
- Design Strategy (DFSS & Circular Economy)
- SMART Engineering & Industry 4.0
- DFMEA/PFMEA, FRACAS & 8D

Strategic Leadership

- Engineering Strategy
- Operational Excellence
- Strategic Planning
- Stakeholder Management
- Risk Management (ERM Framework)

SOFT SKILLS

- Ownership Mindset & Accountability
- Matrix Engineering Leadership
- Change & Transformation Leadership
- Executive Communication & Influence
- Organizational & Pipeline Development
- Structured Problem-Solving

TRAINING

- Design for Six Sigma (DFSS)
- Value Engineering (VAVE)

LANGUAGES

- English: C1 (Advanced)
- German: B2 (In Progress) *Certificate*

INTERESTS

- Biking, Volleyball & Traveling